Final Project Proposal

Project Group Number

11

Group Members

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Project Scope

Chosen Tasks

We plan to complete the following tasks for the final project:

1. **Baseline C/C++ implementation (mandatory)**
2. **OpenMP parallelization and performance analysis**
3. **MPI parallelization and performance analysis**

If time allows, we may also explore a preliminary **MPI+OpenMP hybrid version**, but this is not part of our core scope.

Target Application

Our target application is **PageRank**, a graph analytics algorithm that computes the importance score of each node in a directed graph through iterative updates. We chose this application because it represents a typical sparse and irregular workload, which is highly relevant for performance analysis on modern multicore and distributed-memory systems. It is also a good fit for studying memory access behavior, load imbalance, and communication overhead in parallel implementations.

Initial Assessment of Main Challenges

We expect the main challenges of this project to be the following:

- **Sparse graph representation:** Efficiently storing and traversing the graph using suitable sparse data structures such as CSR/CSC.
- **Irregular memory access:** PageRank involves non-contiguous accesses, which may limit cache efficiency and memory bandwidth utilization.
- **Parallel work distribution:** For OpenMP, the main difficulty is to parallelize the iterative update efficiently while minimizing synchronization overhead.
- **MPI partitioning and communication:** For MPI, we need to design a graph decomposition strategy, manage communication of boundary contributions, and analyze how communication overhead scales with the number of processes.
- **Correctness and convergence:** We need to verify that the parallel versions produce correct PageRank values and converge consistently with the serial baseline.

We expect the **MPI implementation and performance analysis** to require the most effort, especially on multi-node systems, because communication cost and partition quality will strongly affect scalability.

Tentative Schedule

- **April 10 – April 16:**
  Study the reference implementation, prepare input graphs, and implement and verify the serial C/C++ baseline.
- **April 17 – April 23:**
  Perform baseline profiling, identify computational hotspots, select useful performance metrics, and run initial measurements.
- **April 24 – May 2:**
  Implement the OpenMP version, verify correctness, and evaluate speedup and thread scalability.
- **May 3 – May 11:**
  Implement the MPI version, design the graph partitioning and communication strategy, and verify correctness on small and medium graph instances.
- **May 12 – May 20:**
  Run strong-scaling and weak-scaling experiments on Dardel and the school cluster. Analyze communication overhead, scalability, and comparison with the serial and OpenMP versions.
- **May 21 – May 25:**
  If time permits, test a preliminary MPI+OpenMP hybrid configuration. Otherwise, focus on refining experiments, plots, and analysis.
- **May 26 – May 28:**
  Finalize the report, figures, tables, and code packaging.

Division of Responsibilities

Jinye Gong

Responsible for the serial baseline implementation, correctness verification, and the OpenMP version. Also contributes to shared-memory performance evaluation and part of the report writing.

Weiyi Lyu

Responsible for the MPI implementation, distributed-memory experiments on Dardel and the school cluster, and analysis of communication overhead, strong scaling, and weak scaling.

Shared Responsibilities

Both group members will jointly design experiments, discuss performance results, validate correctness, and write and revise the final report together.